# Optimization and Numerical Methods

Weekly assignments

## Assignment 2 (Due on Saturday 15 Feb 2025. Topics: Convex function)

1. Suppose $g : \mathbb{R} \to \mathbb{R}$ is convex, and $c$ and $d$ belongs to the domain of $g$ with $c < d$

   a. Show that
   $$g(x) \leq \frac{d-x}{d-c} g(c) + \frac{x-c}{d-c} g(d)$$
   for all $x \in [c,d]$.

   b. Show that
   $$\frac{g(x)-g(c)}{x-c} \leq \frac{g(d)-g(c)}{d-c} \leq \frac{g(d)-g(x)}{d-x}$$
   for all $x \in (c,d)$. Draw the sketch to illustrate this inequality: determine the slopes of the lines joining the points (for example between $(x, g(x))$, $(c, g(c))$, etc) for each inequalities and compare them with each other.

2. For each of the following functions, determine whether it is convex or concave or neither of them.

   a. $f(x) = e^x - 1$ on $\mathbf{R}$.

   b. $f(x_1, x_2) = 1/(x_1 x_2)$ on $\mathbb{R}^2_{++}$

3. Log-convexity and log-concavity:

   a. Show that the Logistic function $f(x) = e^x/(1+e^x)$ is log-concave with domain of $f = \mathbf{R}$.

   b. Show that the cumulative distribution function of the standard normal distribution is log-concave. Use the condition that a function $f$ is log-concave if and only if $f''(x)f(x) \leq (f'(x))^2$. Additional reference: section 3.5 of Boyd's book.